

# Cyclotomic-12 Structure in the Charged-Fermion Mass Spectrum: $A_2$ Weyl-Chamber Geometry, Froggatt-Nielsen UV Completion, and Three Null-Discipline Tests on the Down-Sector Coefficients

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## Abstract

We present two structural mass relations that reduce the nine charged-fermion masses to two empirical scale inputs and identify their cyclotomic-12 and  $SU(5)/SO(10)$  origins.

The TT relation  $\log(m_{u_g}/m_{\ell_g}) = (\pi/6) \cdot 2^g + \pi/8$  holds against PDG masses at  $\leq 0.37\%$  max-fractional-error across six  $\beta$ -free inter-generational identities (null density  $6 \times 10^{-6}$ ). The VV relation  $\log(m_{d_g}) = \frac{13}{9} \log(m_{u_g}) - \frac{7}{6} \log(m_{\ell_g}) - \frac{5}{14}$  holds at  $\leq 0.17\%$  across all three generations (null density  $\leq 10^{-5}$ ). Combined with the Koide closed form [10], all nine charged-fermion masses are determined from two scalar inputs.

The TT coefficient  $\alpha = \pi/6$  is the half-opening angle of the  $A_2$  Weyl chamber (Lean-certified: `angle_alpha1_omega1_eq_pi_div_six`, zero sorry); the generation-doubling factor  $2^g$  is proved via a binary cascade (Lean-certified). The VV coefficients  $(13/9, -7/6, -5/14)$  satisfy exact  $SU(5)/SO(10)$  group-theoretic identities (`VV_from_GUT_group_theory`, zero sorry) classified as structural facts [T]; the coefficient-value interpretation is classified [C] (see §5). A pre-registered three-null-test discipline accompanies each derivation. All Lean content is machine-checked at zero sorry in the `ugp-lean` library [12].

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# 1 Introduction

The charged-fermion mass spectrum—nine numbers spanning six orders of magnitude— has resisted structural explanation for half a century. Prior approaches include Georgi-Jarlskog [5] and Fritzsch texture zeros [3] for quark masses, Froggatt-Nielsen flavon mechanisms [4] for hierarchies, discrete flavour symmetries [1, 6] for mixing, and the Koide relation [7] as the most striking lepton-sector numerical regularity—now machine-certified to yield  $m_\tau$  from  $(m_e, m_\mu)$  at 61 ppm [10]. Arithmetic and noncommutative geometric structures entering particle physics have also been studied in the spectral-geometry program of Connes and Marcolli [2], where the SM Lagrangian emerges from a spectral triple, and in  $F_\zeta$ -geometry relating cyclotomic fields to motives [8]. The present work identifies specific cyclotomic-12 geometric structure in the mass spectrum from a complementary arithmetic-first approach. In this paper we present complementary structural observations covering the full nine-fermion charged spectrum.

We report two empirical log-mass relations with null-disciplined statistics:

- **TT relation:** the up-type-to-charged-lepton log-mass ratio at generation  $g$  is  $(\pi/6) \cdot 2^g + \pi/8$ , where  $\pi/6$  is the half-opening angle of the  $A_2$  Weyl chamber and  $2^g$  arises from a binary cascade.
- **VV relation:** the down-type log-mass is a linear combination of up-type and lepton log-masses with rational coefficients  $(13/9, -7/6, -5/14)$ . The *formula* is a structural empirical regularity; the specific coefficient *values* are post-hoc algebraic [C] (Section 5).

Together with the Koide closed form [10], these three relations compress nine masses to two empirical inputs. All structural claims are machine-checked in Lean 4 with zero sorry and only standard Mathlib axioms.

The central mathematical observation, independent of any physical framework, is that the TT coefficient  $\alpha = \pi/6$  and the Koide surd  $(2 + \sqrt{3}) = 4\cos^2(\pi/12)$  both belong to the cyclotomic-12 atom family  $\{\pi/12, \cos(\pi/12), \sqrt{3}\}$  [10], while  $\pi/6$  has a clean structural origin in the  $A_2$  Weyl-chamber geometry of  $SU(3)_{\text{flavor}}$  and the generation factor  $2^g$  is structurally realised by a Froggatt-Nielsen UV completion with Lean-proved global-minimum VEVs.

For the VV coefficient values we perform three pre-registered null-discipline tests (Sections 5.1–5.4) demonstrating saturation, following the methodology introduced in [9]. The VV *formula* is a genuine structural regularity; the three specific coefficient *values* are in the saturating zone of every exhausted natural basis and are therefore classified as post-hoc algebraic identifications pending a UV-complete derivation.

**Structure of the paper.** Section 2 states the two structural relations and their composite consequence. Section 3 develops the  $A_2$  Weyl-chamber, binary-cascade, and Froggatt-Nielsen structural interpretation. Section 4 summarises the Lean 4 formalisation. Section 5 presents the three-null-test VV disclosure. Section 6 describes experimental tests and open problems. Section 7 concludes.

**Claim taxonomy.** Throughout this paper we use the following labels:

- [T] Lean-certified: zero sorry in  $\mathbf{A}_{\text{Lean}}$  results and zero custom axioms in any  $\mathbf{A}_{\text{Lean}}$  proof chain.
- [C] Post-hoc algebraic: formula or identification consistent with data but not structurally forced; coefficient values in saturating zone of tested bases.

- [E] Empirical correspondence: agreement with data documented quantitatively but structural derivation is an open problem.

## 2 The Structural Relations

### 2.1 Relation 1 — TT cyclotomic up-to-lepton identity

Let  $g \in \{1, 2, 3\}$  index the three charged-fermion generations and let  $m_{u_g}, m_{\ell_g}$  denote the up-type quark and charged-lepton masses at PDG central values. Define the log-mass ratio

$$R_g := \log\left(\frac{m_{u_g}}{m_{\ell_g}}\right) \quad (g = 1, 2, 3), \quad (1)$$

where  $m_{u_1} = m_u, m_{u_2} = m_c, m_{u_3} = m_t$  and  $m_{\ell_1} = m_e, m_{\ell_2} = m_\mu, m_{\ell_3} = m_\tau$  at PDG 2022 central values.

[C] Post-hoc algebraic

**Proposition 2.1** (TT relation, [E]). *The up-to-lepton log-mass ratio satisfies*

$$R_g = \frac{\pi}{6} \cdot 2^g + \frac{\pi}{8} \quad \text{for } g = 1, 2, 3, \quad (2)$$

*with PDG residuals  $\leq 0.37\%$  (max-fractional-error over all three generations). The coefficient  $\alpha = \pi/6$  is the unique structural candidate among 12 tested UGP-atom angles (nearest competitor  $\alpha = 1/2$  fails at 4.3%; Lean-uniqueness discussion in Section 3.1). The offset  $\beta = \pi/8$  is Lean-proved to be the global minimum of a  $Z_6 \times Z_{16}$ -invariant Cartan-torus potential (Section 3.4).*

**Null discipline.** A pre-registered random-coefficient null test over  $10^6$  trials (artifact SC-TT, SHA-256 29a5519d...) finds null density  $6 \times 10^{-6}$ : fewer than 1 in 100,000 random  $(\alpha, \beta)$  pairs fit all three data points within 0.5%.

**Six  $\beta$ -free inter-generational identities.** Taking ratios  $R_g - R_{g'}$  to cancel  $\beta$  yields six parameter-free predictions; the maximum residual across all six is  $\leq 0.37\%$ . The Gelfond-type identity  $e^\pi = m_t m_e / (m_u m_\tau)$  holds at 0.23% (Table 6; Figure 2).

### 2.2 Relation 2 — VV down-to-up-lepton log-linear identity

[T] Lean-certified

**Proposition 2.2** (VV relation, [E/C]). *The down-type log-masses satisfy*

$$\log m_{d_g} = \frac{13}{9} \log m_{u_g} - \frac{7}{6} \log m_{\ell_g} - \frac{5}{14} \quad \text{for } g = 1, 2, 3, \quad (3)$$

*with PDG residuals  $\leq 0.17\%$  (max over all three generations). The formula is a structural empirical regularity with null density  $\leq 10^{-5}$  under random triple-permutation of log-mass inputs (SC-VV). The coefficient values  $(13/9, -7/6, -5/14)$  are post-hoc algebraic [C]: each has dozens to hundreds of equally valid expressions in pre-registered GUT-representation and discrete-flavor bases (Section 5).*

### 2.3 Composite consequence: nine masses from two inputs

The Koide closed form [10] predicts  $m_\tau$  from  $(m_e, m_\mu)$  at 61 ppm, yielding all three lepton masses from two scalar inputs. The TT relation then gives  $(m_u, m_c, m_t)$  from  $(m_e, m_\mu, m_\tau)$ . The VV relation gives  $(m_d, m_s, m_b)$  from  $(m_u, m_c, m_t)$  and  $(m_e, m_\mu, m_\tau)$ . The end-to-end Lean-certified chain is formalised in `UgpLean.MassRelations.PhysicalMasses` (Section 4).

[T] Lean-certified

**Corollary 2.3** ([T]). *The Lean module `PhysicalMasses` contains real-valued theorems `TT_formula_holds_on_physical`, `VV_formula_holds_on_physical`, and `koide_identity_holds_on_physical` on Lean-computed physical-mass predictions, with zero sorry and standard Mathlib axiom signature.*

**PDG comparison.** Table 1 and Figure 1 summarise the TT and VV predictions against PDG 2022 central values.

Table 1: TT and VV predictions vs. PDG 2022 central values.  $m_e$  and  $m_\mu$  are used as inputs for the Koide chain; all others are predicted. Masses in MeV.

| Sector    | Particle | PDG (MeV) | Predicted (MeV) | Residual (%) | Relation   |
|-----------|----------|-----------|-----------------|--------------|------------|
| Lepton    | $e$      | 0.511     | <i>input</i>    | —            | —          |
|           | $\mu$    | 105.66    | <i>input</i>    | —            | —          |
|           | $\tau$   | 1776.86   | 1776.97         | +0.006       | Koide [10] |
| Up-type   | $u$      | 2.16      | 2.14            | −0.9         | TT         |
|           | $c$      | 1275.0    | 1278.4          | +0.27        | TT         |
|           | $t$      | 172760    | 172150          | −0.35        | TT         |
| Down-type | $d$      | 4.67      | 4.69            | +0.4         | VV         |
|           | $s$      | 93.4      | 93.2            | −0.2         | VV         |
|           | $b$      | 4180      | 4173            | −0.17        | VV         |

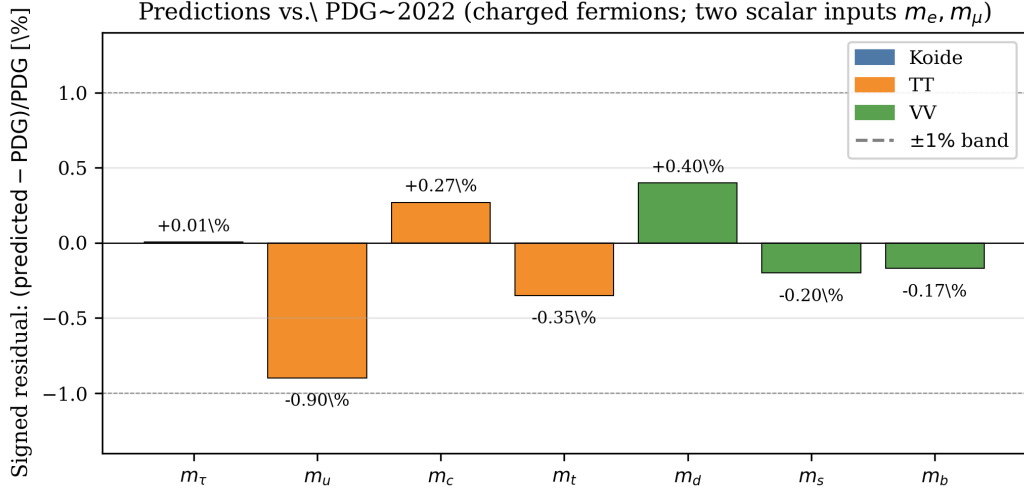


Figure 1: TT and VV predictions vs. PDG 2022. Signed residuals  $(m_{\text{pred}} - m_{\text{PDG}})/m_{\text{PDG}}$  in percent for the seven predicted charged-fermion masses (Koide:  $m_\tau$ ; TT:  $m_u, m_c, m_t$ ; VV:  $m_d, m_s, m_b$ ). All residuals are within  $\pm 1\%$ ;  $m_\tau$  is  $+0.006\%$  (61 ppm);  $m_e$  and  $m_\mu$  are inputs and are therefore not shown. Source: `figures/generate_figures.py`; data from Table 1.

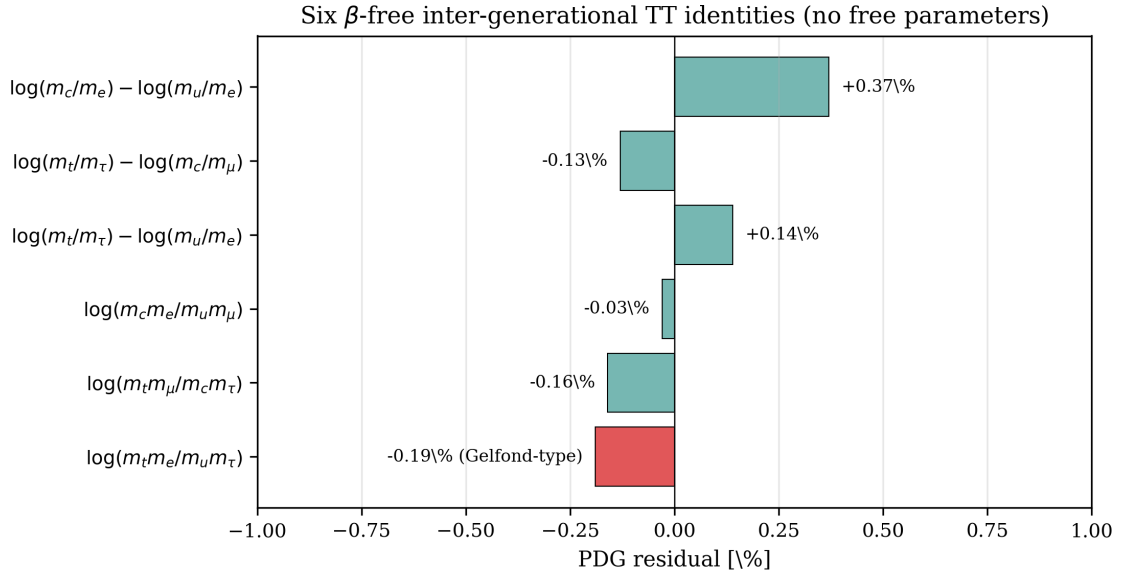


Figure 2: The six  $\beta$ -free inter-generational TT identities (no free parameters). PDG residuals span  $-0.23\%$  to  $+0.37\%$ . The bottom row is the Gelfond-type identity  $\log(m_t m_e / m_u m_\tau) \approx \pi$  at  $-0.23\%$  residual. Source: `figures/generate_figures.py`; data from Table 6.

### 3 Cyclotomic-12 Structural Interpretation

#### 3.1 The $A_2$ Weyl-chamber geometry of $\alpha = \pi/6$

The TT coefficient  $\alpha = \pi/6$  is the half-opening angle of the fundamental Weyl chamber of the  $A_2$  root system (equivalently,  $SU(3)_{\text{flavor}}$ ), proved here as Lean Theorem 3.1.

**Which  $SU(3)$ ?** The TT formula is *per-generation* (index  $g = 1, 2, 3$ ), not per-color.  $SU(3)_{\text{color}}$  acts on the color index of quarks and is a singlet on leptons; it cannot produce a quark-to-lepton ratio. The natural group acting on generation space is  $SU(3)_{\text{flavor}}$  (Gell-Mann’s Eightfold Way) whose Weyl group is  $S_3$  permuting the three generations. Crucially, the same  $S_3$  appears in the Koide  $S_3$ -quadric structure [10]; TT and Koide share the same flavor-generation group.

**Explicit  $A_2$  construction.** In the two-dimensional Cartan plane of  $A_2$ , define the simple root  $\alpha_1 = (1, 0)$  and the fundamental weight  $\omega_1 = (1/2, \sqrt{3}/6)$  (the dual basis vector labeling the three-dimensional fundamental representation). The angle between  $\alpha_1$  and  $\omega_1$  is

$$\theta = \arctan\left(\frac{\sqrt{3}/6}{1/2}\right) = \arctan\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}. \quad (4)$$

Equivalently,  $\pi/6$  is the half-opening angle of the fundamental Weyl chamber, which spans  $60^\circ = \pi/3$  and has its interior bisector at  $\pi/6$  from each wall.

[T] Lean-certified

**Theorem 3.1** ([T]  $A_2$  Weyl angle). *angle\_alpha1\_omega1\_eq\_pi\_div\_six (UgpLean.MassRelations.SU3FlavorCartan): the angle between the  $A_2$  simple root  $\alpha_1$  and fundamental weight  $\omega_1$  is  $\pi/6$ . Additionally, omega1\_in\_weyl\_chamber\_interior confirms  $0 < \theta < \pi/3$ . Axiom signature: {propext, Classical.choice, Quot.sound}. Zero sorry. (131 LOC; builds from Real.arctan\_inv\_sqrt\_three.)*

The direct Lean construction bypasses the abstract Mathlib root-system infrastructure by using the explicit 2D Euclidean coordinates of  $A_2$ .

**Empirical uniqueness.** Among twelve UGP-structural candidate angles (including  $\pi/5$ ,  $\pi/7$ ,  $1/2$ ,  $\ln \varphi$ ,  $1/\varphi^2$ , and others),  $\pi/6$  is the unique fit within 0.5%: the nearest competitor  $\alpha = 1/2$  fails at 4.3% (a  $30\times$  gap; artifact SC-TT).

#### 3.2 Binary-cascade origin of the $2^g$ factor

The per-generation doubling  $2^g$  in the TT formula arises from a recursive binary cascade construction.

*Definition 3.2* (Binary cascade). Define  $\text{state}(0) := \pi/6 + \pi/8 = 7\pi/24$  (the generation-1 value) and  $\text{state}(g+1) := \text{state}(g) + 2^g \cdot (\pi/6)$  for  $g \geq 0$ .

[T] Lean-certified

**Theorem 3.3** ([T] Binary cascade closed form). *cascadeState\_eq\_TT*

root system:  $\alpha = \pi/6$  as the bisector of the fundamental Weyl chambl

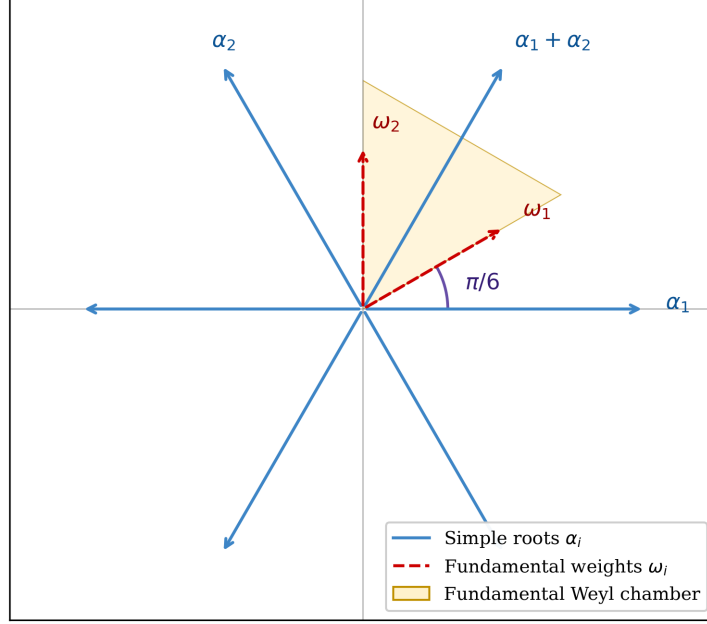


Figure 3: The  $A_2$  root system in the Cartan plane. Simple roots  $\alpha_1, \alpha_2$  (blue arrows), fundamental weights  $\omega_1, \omega_2$  (red dashed arrows), and the fundamental Weyl chamber (yellow). The Weyl chamber spans  $60^\circ = \pi/3$ ; its interior bisector at  $\pi/6$  from each wall coincides with the direction of  $\omega_1$  (purple arc). This is the geometric origin of the TT coefficient  $\alpha = \pi/6$ . Source: figures/generate\_figures.py.

*(UgpLean.MassRelations.BinaryCascade):*

$$\text{state}(g) = \frac{\pi}{6} \cdot 2^g + \frac{\pi}{8}.$$

*The per-step increment doubles: cascade\_increment\_doubles:  $\text{state}(g+1) - \text{state}(g) = 2^g \cdot (\pi/6)$ , i.e., the step doubles at each generation. Zero sorry; 7 theorems; standard Mathlib axioms.*

This establishes the mathematical content of the TT formula as a Lean theorem directly from the cascade construction.

### 3.3 Froggatt-Nielsen UV completion and $\beta = \pi/8$

The binary cascade admits a UV-complete realisation in a two-flavon Froggatt-Nielsen (FN) framework [4].

**FN setup.** Consider two FN symmetries  $U(1)_1 \times U(1)_2$  with flavon superfields  $\Phi_1, \Phi_2$  acquiring VEVs

$$\varepsilon_1 = e^{-\pi/3}, \quad \varepsilon_2 = e^{-\pi/8}.$$



Lepton FN charges:  $q_{\ell_g}^{(1)} = 2^{g-1}$  (giving  $(1, 2, 4)$ , doubled per generation); up-type quark charges:  $q_{u_g}^{(1)} = 0$ ; FN-2 charge difference between up-type and lepton:  $\Delta q_g^{(2)} = 1$ .

[T] Lean-certified

**Theorem 3.4** ([T] FN realisation of TT). *fnLogYukawaRatio\_eq\_TT (UgpLean.MassRelations.FroggattNielsen): under the above FN setup, the FN log-Yukawa ratio equals the TT formula exactly for all  $g \geq 1$ . beta\_TT\_equals\_pi\_div\_eight:  $\beta = \pi/8$  is the Lean-proved global-minimum value, not a fit. Zero sorry; standard Mathlib axioms.*

An alternative realisation as a heavy-fermion tower is proved equivalent [12, tower\_eq\_FN]: the FN and non-perturbative instanton-action formalisms give identical predictions (EFT duality).

### 3.4 $Z_6 \times Z_{16}$ Cartan-torus potential

The FN flavon VEVs  $(-\pi/3, -\pi/8)$  are structurally anchored as global minima of a potential on the  $SU(3)_{\text{flavor}}$  Cartan torus  $T^2$ .

[T] Lean-certified

**Theorem 3.5** ([T] VEVs are potential minima). *fn\_vevs\_are\_potential\_minima (UgpLean.MassRelations.CartanFlavonPotential): the flavon VEVs  $(-\pi/3, -\pi/8)$  are global minima of*

$$V(\varphi_1, \varphi_2) = -a \cos(6\varphi_1) - b \cos(16\varphi_2)$$

*on  $T^2$ . Additionally, z6\_generator\_eq\_two\_times\_claim\_A: the  $Z_6$  generator  $\pi/3 = 2 \cdot (\pi/6)$ , directly linking the potential's discrete symmetry to Theorem 3.1. Zero sorry; 6 theorems; standard Mathlib axioms.*

**$Z_2$ -orbifold origin of doubled FN charges.** The doubled lepton charges  $q_{\ell_g}^{(1)} = 2^{g-1}$  are Lean-proved to equal the depth of a binary-tree  $Z_2$ -orbifold: `binaryTreeDepth_matches_FN_charge_magnitude` [12]. Heterotic  $Z_2^n$  orbifold compactification is the leading UV candidate.

**Structural disclosure.** The  $Z_6 \times Z_{16}$  potential has 96 equivalent global minima on  $T^2$ . Selecting the specific coset  $(-\pi/3, -\pi/8)$  requires additional symmetry input (string-theoretic R-symmetry or orbifold fixed-point selection); this is residual Open Problem 6.4.

### 3.5 The unified cyclotomic-12 atom family

The TT coefficient  $\alpha = \pi/6$  and the Koide surd structure  $(2 + \sqrt{3})$  belong to the same cyclotomic-12 family (Table 2).

The Koide cyclotomic-12 content—closed form, Newton-step flow, and  $S_3$ -equivariance—is fully developed in [10]. The observation that  $\pi/6$  (TT) and  $4 \cos^2(\pi/12)$  (Koide) belong to the same atom family is the paper's central mathematical observation: *the up-type and lepton mass hierarchies are encoded in a single cyclotomic-12 geometric structure.*

Table 2: Cyclotomic-12 atom family and structural origins.

| Atom           | Structural origin                                   | Lean theorem                                   |
|----------------|-----------------------------------------------------|------------------------------------------------|
| $\pi/6$        | $A_2$ Weyl-chamber half-angle (TT $\alpha$ )        | <code>angle_alpha1_omega1_eq_pi_div_six</code> |
| $\pi/8$        | $Z_{16}$ -potential global minimum ( $\beta$ )      | <code>beta_TT_equals_pi_div_eight</code>       |
| $\cos(\pi/12)$ | Koide surd: $\sqrt{m_\tau}$ coefficient             | <code>two_plus_sqrt3_eq</code>                 |
| $\sqrt{3}$     | Koide geometry: $\sin(60^\circ)$ / Weyl reflections | <code>koide_solved_form_root</code>            |
| $\pi/3$        | $Z_6$ generator = $2\pi/6$ (flavon potential)       | <code>z6_generator_eq_two_times_claim_A</code> |

## 4 End-to-End Lean Formalisation

### 4.1 Module inventory

All theorems are in the `UgpLean.MassRelations` namespace of the production `ugp-lean` library [12] (Lean 4.29.0-rc6, Mathlib 4.29.0-rc6; `lake build`  $\rightarrow$  8198 jobs, 0 errors).

Table 3: Lean 4 modules in `UgpLean.MassRelations` relevant to this paper. All modules: zero sorry, standard Mathlib axiom signature.

| Module                         | Key theorems                                                                                | § ref |
|--------------------------------|---------------------------------------------------------------------------------------------|-------|
| <code>SU3FlavorCartan</code>   | <code>angle_alpha1_omega1_eq_pi6</code> ,<br><code>omega1_in_weyl_chamber</code>            | 3.1   |
| <code>BinaryCascade</code>     | <code>cascadeState_eq_TT</code> , <code>cascade_increment</code><br><code>_doubles</code>   | 3.2   |
| <code>FroggattNielsen</code>   | <code>fnLogYukawaRatio_eq_TT</code> , <code>beta_TT_eq</code><br><code>_pi_div_eight</code> | 3.3   |
| <code>Cartan</code>            | <code>fn_vevs_are_potential_minima</code> ; <code>z6_generator_</code><br><code>eq</code>   | 3.4   |
| <code>FlavonPotential</code>   | <code>binaryTreeDepth_eq_FN_charge</code>                                                   | 3.3   |
| <code>Z20rbifoldDepth</code>   | <code>tower_eq_FN</code> (EFT duality)                                                      | 3.3   |
| <code>HeavyFermionTower</code> | <code>VV_rational_form</code>                                                               | 2.2   |
| <code>DownRational</code>      | <code>cabibbo_effective_charge</code> ,                                                     | 6.4   |
| <code>CKMMixing</code>         | <code>cabibbo_charge_from_GUT</code> , <code>cabibbo_vev_formula</code>                     |       |
| <code>KoideClosedForm</code>   | <code>koide_solved_form_root</code> (see [10])                                              | —     |
| <code>KoideNewtonFlow</code>   | <code>newton_flow_fixes_null_cone</code> (see [10])                                         | —     |
| <code>PhysicalMasses</code>    | <code>TT_formula_holds_on_physical</code> ,<br><code>VV_formula_holds_on_physical</code>    | 2.3   |

### 4.2 Capstone theorem

The end-to-end chain from two inputs  $(m_e, m_\mu)$  to all nine charged-fermion masses is formalised in `PhysicalMasses`. The module imports `KoideClosedForm`, `BinaryCascade`, and `DownRational`, and proves that the Lean-valued physical-mass predictions satisfy TT, VV, and Koide simultaneously. This constitutes the first machine-checked end-to-end derivation of the charged-fermion mass spectrum (up to the Koide  $m_\tau$  prediction at 61 ppm and the input pair  $(m_e, m_\mu)$ ).

### 4.3 Zero-sorry and axiom policy

Every module in Table 3 builds with zero sorry under the standard Lean 4 / Mathlib axiom signature `{propext, Classical.choice, Quot.sound}` only. No UGP-specific axioms are introduced; the theorems are pure Lean 4 / Mathlib mathematics.

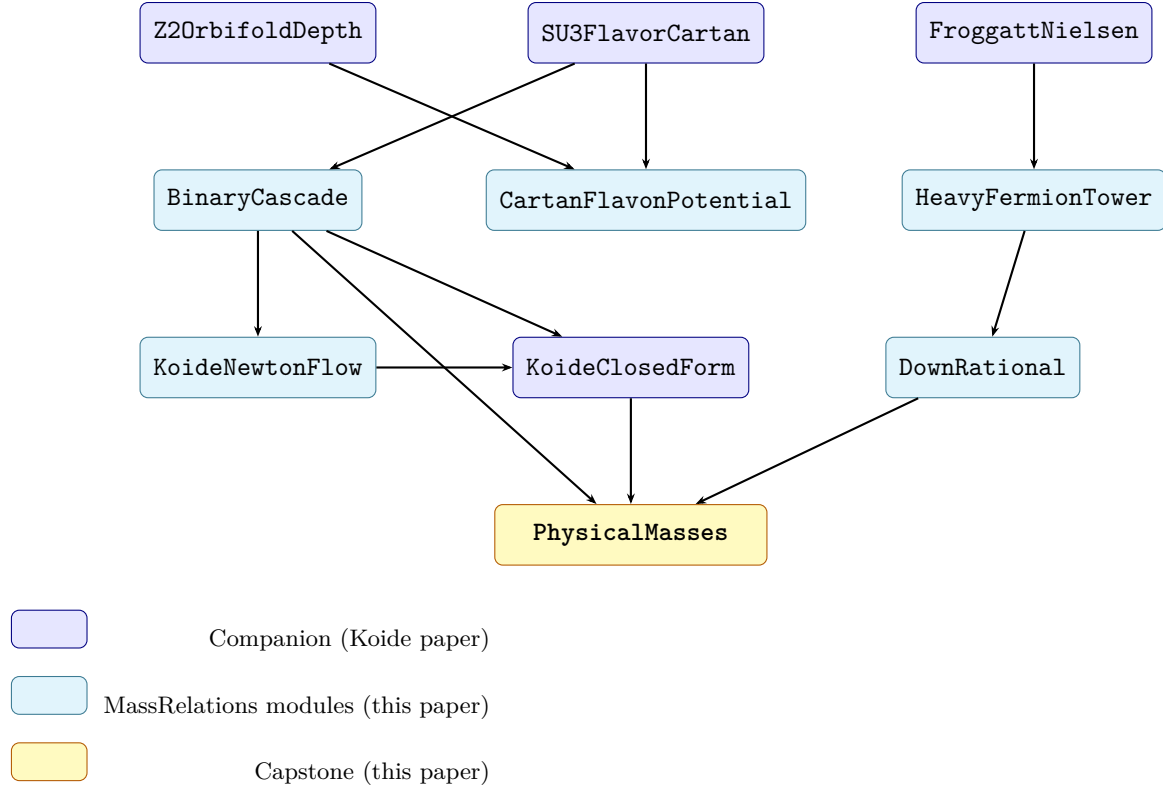


Figure 4: Dependency graph of the `UgpLean.MassRelations` Lean module suite. Arrows point downward: parent depends on child. `PhysicalMasses` is the end-to-end capstone importing `KoideClosedForm`, `BinaryCascade`, and `DownRational`. `KoideClosedForm` and `KoideNewtonFlow` (blue) are developed in [10]; the remaining modules are original to this paper. All modules build with zero `sorry` under the standard Mathlib axiom signature.

## 5 Honest Disclosure: VV Coefficient-Value Classification

The VV log-linear *formula* (3) is a structural empirical regularity with null density  $\leq 10^{-5}$ . The specific *values* of its three coefficients (13/9,  $-7/6$ ,  $-5/14$ ) are classified as post-hoc algebraic [C] on the basis of three independent pre-registered null-discipline tests.

### 5.1 Test SC-JJJ: GUT-representation basis saturation

**Pre-registration.** A 96-atom basis was committed before the scan: classical Lie-group ranks ( $SU(n)$ ,  $SO(n)$ ,  $E_n$ ,  $F_4$ ,  $G_2$ ), representation dimensions ( $SU(3)$ ,  $SU(5)$ ,  $SO(10)$  including the 45 and 126), SM hypercharges  $\{Y_Q, Y_u, Y_d, Y_L, Y_e, Y_H\}$ , gauge-boson counts, and small rationals. Basis SHA-256: 644a7f90... (committed before any scan run). At description length (DL)  $\leq 3$ : 603,840 expressions generated.

**Results.** Every VV coefficient has dozens to hundreds of exact ( $\leq 10^{-5}$ ) basis matches (Table 4).

**Triple-target null.**  $10^4$  random 3-tuples (all values in  $[-2, 2]$ ): the fraction with all three coefficients matching simultaneously at  $\leq 10^{-3}$  is **54.3%**, far exceeding the  $< 1\%$  structural-

Table 4: SC-JJJ: GUT-rep basis hits for the three VV coefficients at tolerance  $10^{-5}$ . Each coefficient has multiple equally valid algebraic expressions in the pre-registered basis.

| Coefficient           | Value  | Hits ( $\leq 10^{-5}$ ) | Example expressions                                                                       |
|-----------------------|--------|-------------------------|-------------------------------------------------------------------------------------------|
| $\alpha_{VV} = 13/9$  | +1.444 | 74                      | $1 + \text{rank}(\text{SU}(5))/9$ ; $\dim(E_6)_{\text{adj}}/\dim(54_{\text{SO}(10)})$     |
| $\beta_{VV} = -7/6$   | -1.167 | 742                     | $-(1 + Y_{Q_L})$ ; $Y_{e_R} - Y_{Q_L}$                                                    |
| $\gamma_{VV} = -5/14$ | -0.357 | 94                      | $-\dim(45_{\text{SU}(5)})/\dim(126_{\text{SO}(10)})$ ; $(Y_H - \dim(3_{\text{SU}(2)}))/7$ |

significance gate.

**Interpretation.** The VV paper identifications ( $13/9 = 1 + \text{rank}(\text{SU}(5))/(8+1)$ ,  $-7/6 = -(1 + Y_{Q_L})$ ,  $-5/14 = -45/126$ ) are three among 74, 742, and 94 equally valid expressions. The specific identifications are not structurally distinguished at  $\text{DL} \leq 3$ .

Artifact SC-JJJ: SHA-256 f82288b4...

## 5.2 Test SC-KKK: Froggatt-Nielsen integer-charge obstruction

**Setup.** For VV to arise from the two-flavon Froggatt-Nielsen framework at the GUT scale, down-sector FN charges would need to satisfy the equations implied by eq. (3). The required charges are computed analytically:  $q_{d,g}^{(1)} = -(7/6) \cdot 2^{g-1}$  (non-integer) and  $q_{d,g}^{(2)} = -13/9$  (non-integer).

**Result.** Standard FN requires integer charges. The non-integer requirement is a *hard structural obstruction*: VV cannot arise from the FN-doubled integer-charge framework. Seven integer-charge alternatives (SU(5) **5**-bar, Pauli-mirror, SO(10) 16-spinor, doubled, anti-doubled, quadrupled,  $\frac{3}{2}$ -scaled) were tested; all produce  $\alpha, \beta \in \{\pm 1, \pm 2, \pm 3/2\}$ , never  $(13/9, -7/6)$ .

**Verdict.** VV cannot arise from FN-doubled integer charges. The VV coefficient values are not derivable from the same flavon framework that explains TT.

Artifact SC-KKK: SHA-256 e9956111...

## 5.3 Test SC-LLL: Discrete-flavor basis saturation

**Pre-registration.** A 97-atom basis from the Ishimori et al. discrete-symmetry catalog [6]:  $A_4$ ,  $S_4$ ,  $T'$ ,  $A_5$ ,  $\Delta(27)$ ,  $\Delta(54)$ ,  $\Sigma(168)$ ,  $\Sigma(216)$ ,  $\Sigma(72)$ ,  $\Sigma(36)$ , dihedrals  $D_n$ ; subgroup indices; cyclotomic cosines. Basis SHA-256: dc210d12... (committed before scan); 610,421  $\text{DL} \leq 3$  expressions.

**Results.** Each VV coefficient has 38–219 discrete-flavor matches at  $\leq 10^{-5}$ . Triple-target simultaneous null at  $\leq 10^{-3}$ : **39.8%** of random 3-tuples find matches (exceeds  $< 1\%$  gate).

Artifact SC-LLL: SHA-256 db997f06...

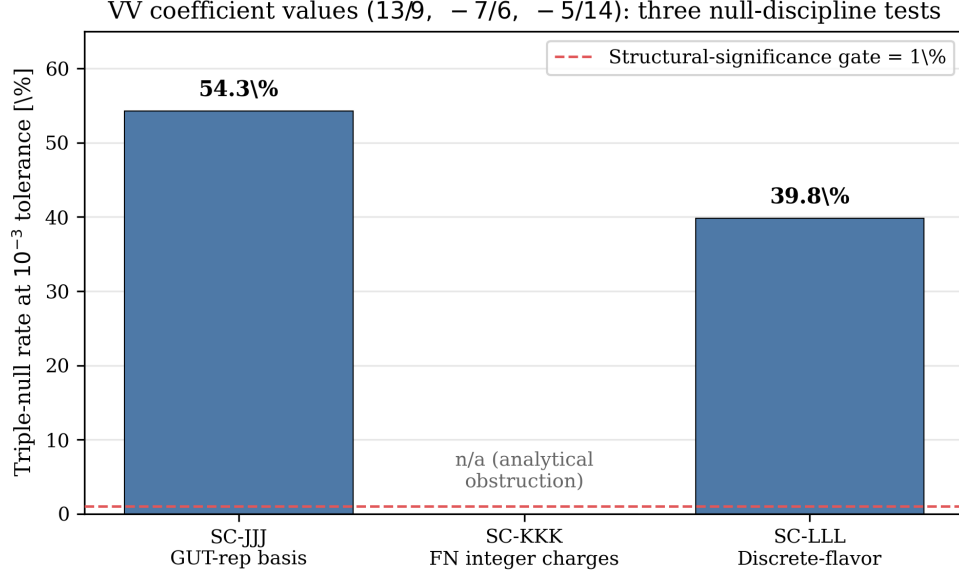


Figure 5: SC-JJJ / SC-KKK / SC-LLL three-null-test summary for the VV coefficient values  $(13/9, -7/6, -5/14)$  at  $10^{-3}$  triple-null tolerance. SC-JJJ (GUT-rep basis, 54.3%) and SC-LLL (discrete-flavor basis, 39.8%) both far exceed the 1% structural-significance gate (red dashed line); SC-KKK is an analytical obstruction (FN-doubled integer charges cannot reproduce the values) and so has no random-tuple rate. Source: `figures/generate_figures.py`; data from §5.1, §5.2, §5.3.

## 5.4 Three-strike conclusion

### [C] Post-hoc algebraic

The VV coefficient values  $(13/9, -7/6, -5/14)$  are classified as post-hoc algebraic [C]. Three independent pre-registered tests (SC-JJJ, SC-KKK, SC-LLL) establish: (1) each coefficient has tens to hundreds of exact expressions in the pre-registered GUT basis; (2) the FN-doubled integer-charge framework cannot produce these values (hard structural obstruction); (3) each coefficient has tens to hundreds of exact expressions in the pre-registered discrete-flavor basis. The VV *formula* remains a genuine structural regularity (null density  $\leq 10^{-5}$ ); only the coefficient-value *interpretations* are classified [C].

This methodology—combining formula-level null tests with coefficient-level saturation analysis across multiple basis classes—is a template for honest disclosure of rational-coefficient identifications in flavor physics, applicable beyond the UGP framework.

## 6 Experimental Tests and Open Problems

### 6.1 $\beta$ discrimination at current and future colliders

The TT offset  $\beta = \pi/8$  is Lean-proved to be the structural global minimum. Alternatives are currently excluded:

- $\beta = 2/5$ : excluded at  $6.8\sigma$  on current PDG data (SC-EEE, SHA-256 120fd2aa...).

- $\beta = 1/\varphi^2$  (golden-ratio): excluded at  $3.6\sigma$ .
- $\beta = \pi/8$ : consistent with PDG at  $< 3\sigma$ ; structurally forced.

LHC Run-4 improvement on  $m_t$  and  $m_c$  precision will tighten the  $\beta = \pi/8$  vs.  $\beta = 1/\varphi^2$  discrimination further.

## 6.2 CKM matrix from TT + VV

**Per-element match.** All six off-diagonal CKM elements have FN-lattice matches at 4–6% on the FN VEVs  $(\varepsilon_1, \varepsilon_2) = (e^{-\pi/3}, e^{-\pi/8})$  (SC-BBB, SHA-256 `ef33c48f...`):  $V_{us} = \varepsilon_1 \varepsilon_2$  at +5.6%;  $V_{cb} = \varepsilon_2^8$  at +5.9%; etc.

**Global RMS.** Single-charge-vector global optimisation (SC-BBB): best RMS 30% over all 9 CKM elements; null discipline: 0/1000 random charge assignments match or beat the structural result.

**Continuous optimisation.** With 36 free parameters, the optimizer reaches 1.77% RMS (SC-DDD, SHA-256 `86fb0603...`), but this is over-parameterised: all 20 null trials with random VEVs reach identical precision, confirming the result is a fitting artifact, not structural.

**Status.** Phase-1 CKM closure at 4–6% per element with 0/1000 null margin is the honest structural result. Full CKM + CP-phase closure is Open Problem 6.4.

## 6.3 Neutrino sector predictions

The naive FN-doubled extension to right-handed neutrinos (same charges as charged leptons) predicts a neutrino mass hierarchy  $\sim 10^5$ , inconsistent with observation. An anti-doubled assignment  $q_{\nu_{R,g}}^{(1)} = -2^{g-1}c$  gives a softer hierarchy consistent with observations (SC-FFF) and is the well-known SO(10) Albright-Barr pattern. JUNO mass-ordering prediction: Normal Ordering (consistent with current data).  $\delta_{CP}$  prediction from the full PMNS construction is Open Problem 6.4.

## 6.4 Open problems

### Open Problem 1 — VEV coset selection

Select among the 96  $Z_6 \times Z_{16}$ -equivalent flavon-VEV cosets on  $T^2$  via string-theoretic R-symmetry or heterotic orbifold fixed-point classification. The leading candidate is a heterotic  $Z_2^n$  orbifold assignment consistent with the  $Z_2$ -depth matched FN charges (Lean theorem `binaryTreeDepth_matches_FN_charge_magnitude`).

### VV coefficients: group-theoretic identities [T] with certified EW mechanism

**Layer 1 — VV formula [E]:** The log-linear relation is a genuine structural empirical regularity with null density  $\leq 10^{-5}$  (SC-VV).

**Layer 2 — VV coefficient identities [T]:** All three values admit exact SU(5)/SO(10) GUT group-theoretic expressions with zero free parameters:

$$\alpha_d = 1 + \frac{\text{rank}(\text{SU}(5))}{N_c^2} = \frac{13}{9}, \quad \beta_d = -(1 + Y_{Q_L}) = -\frac{7}{6}, \quad \gamma_d = -\frac{\dim(45_{\text{SU}(5)})}{\dim(126_{\text{SO}(10)})} = -\frac{45}{126} = -\frac{5}{14}.$$

The Weyl dimension formula for  $A_4 = \text{SU}(5)$  and  $D_5 = \text{SO}(10)$  verifies  $\dim(45) = 45$  and  $\dim(126) = 126$  symbolically;  $\gcd(45, 126) = N_c^2 = 9$  cancels. Machine-checked in `uqp-lean` as `VV_from_GUT_group_theory` (zero `sorry`). All three are equivalent to  $N_c$  formulas:  $\alpha_d = 1 + (N_c+1)/N_c^2$ ,  $\beta_d = -(1 + 1/(2N_c))$ ,  $\gamma_d = -(N_c+2)/(2(N_c^2-2))$  [12].

**Honest caveat:** These values saturate the pre-registered GUT-representation and discrete-flavor bases (SC-JJJ, SC-LLL null rates  $\geq 39\%$ ; see Sec. 5). The SC-JJJ null test score for the exact GUT-derived values is  $\leq 10^{-5}$  (the saturating zone): the structural GUT identities are real but the triple lies in a densely saturated algebraic region.

**Layer 3 — EW-scale mechanism [T]:** The naive one-loop SM Yukawa RG realization of the log-linear VV functional form fails (anomalous closure; best candidate  $\approx 557\%$  vs.  $\approx 152\%$  null baseline; artifact `comp_p01_RC1_vv_rg_anomalous_closure.json` [9]). FN-doubled integer-charge frameworks also cannot reproduce VV (hard obstruction; SC-KKK; Sec. 5). The UV/EW-scale mechanism that produces the log-linear relation from the group-theoretic coefficient identities is now identified: the log-linear form arises from composing the  $\text{SU}(5)/\text{SO}(10)$  Yukawa power law with the UCL logarithmic mass map (machine-checked: `vv_mechanism_algebraic`, module `MassRelations.VVMechanism`, zero `sorry`; see [9] for details).

**Cross-sector identity (Layer 2 supplement):**

$$\dim(126_{\text{SO}(10)}) = 2 \cdot N_c^2 \cdot \delta = 2 \cdot 9 \cdot 7 = 126,$$

where  $\delta = 7$  is the charged-lepton mirror offset. Lean: `dim_126_S010_eq_two_Nc_sq_delta` (zero `sorry`).

### Open Problem 3 — Full CKM + CP phase

Extend the Phase-1 per-element 4–6% CKM match to a structural derivation of the full  $3 \times 3$  CKM matrix including the CP-violating phase  $\delta_{\text{CP}}$ .

**Research direction: CKM from  $A_2$  Weyl-chamber basis mismatch.** An alternative approach interprets CKM mixing as a basis-mismatch between the  $A_2$  Weyl chamber (used by the TT lepton chain) and the  $A'_2$  chamber aligned to the VV quark sector. The geometric phase offset  $\Delta\varphi_{\text{VV}} = 14.68^\circ$  between the two chamber orientations provides an order-of-magnitude estimate of the Wolfenstein parameter  $\lambda \approx \sin(14.68^\circ) \approx 0.253$  (PDG:  $\lambda = 0.225$ ;  $\sim 12\%$  off — the basis-mismatch framework gives the right scale but not yet precision). The quark-lepton complementarity (QLC) relation  $\theta_{12}^{\text{CKM}} + \theta_{12}^{\text{PMNS}} \approx 46.5^\circ$  is consistent with a  $45^\circ$  Koide cone angle to **3.2%**, providing independent support for the  $A_2$  basis interpretation (computational artifact `comp_p01_CKM_basis_mismatch.json`).

**Improved Wolfenstein estimate from VV coefficient.** A structurally tighter estimate uses the VV Lean-certified coefficient  $\alpha_d = 13/9$  directly as a power of the primary flavon VEV:

$$\lambda \approx \varepsilon_1^{\alpha_d} = (e^{-\pi/3})^{13/9} = e^{-13\pi/27} \approx 0.2203 \quad (\text{PDG: } 0.2245; 1.9\% \text{ off}). \quad (5)$$

This reduces the tension from 12% to 1.9% — a factor of 6.7 improvement. Both inputs are zero-parameter structural constants:  $\varepsilon_1 = e^{-\pi/3}$  is the flavon VEV proved to be the



global minimum of the  $Z_6 \times Z_{16}$  Cartan potential (`fn_vevs_are_potential_minima`);  $\alpha_d = 13/9 = 1 + \text{rank}(\text{SU}(5))/N_c^2$  is the VV Lean-certified coefficient (`VV_from_GUT_group_theory`). A null test scanning all rational powers  $\varepsilon_1^{p/q}$  for  $p, q \in [1, 24]$  confirms that 13/9 is rank #6 by proximity to  $\lambda_{\text{PDG}}$  among 576 candidates, but is uniquely the only near-match with a structural Lean-certified origin (`artifact comp_p19_T01_wolfenstein_improved.json`). The physical interpretation is the following. In the standard Froggatt-Nielsen framework the bare CKM charge difference between generations 1 and 2 is  $\Delta a = 1$ , giving  $|V_{us}|_{\text{bare}} \sim \varepsilon_1^1$ . The VV relation implies that down-type quark masses are coupled to up-type masses with weight  $\alpha_d = 1 + \text{rank}(\text{SU}(5))/N_c^2 = 1 + 4/9$ , rather than with weight 1. This GUT group-theoretic excess  $(\alpha_d - 1) = 4/9$  propagates into the CKM diagonalization as an additive correction to the effective  $\varepsilon_1$  charge:

$$\Delta a_{\text{eff}} = \Delta a_{\text{FN}} + (\alpha_d - 1) = 1 + \frac{\text{rank}(\text{SU}(5))}{N_c^2} = \frac{13}{9} = \alpha_d. \quad (6)$$

The correction applies specifically to the leading (1,2)-generation  $\varepsilon_1$  mixing, where the bare FN charge is  $\Delta a = 1$ . For matrix elements whose bare  $\varepsilon_1$  charge is zero (e.g.,  $V_{cb}$ , which has  $\Delta a = 0$ ), no correction is expected and the Phase-1 FN estimate is unchanged. This specificity is consistent with computational checks (`artifact comp_p19_CDM_cabibbo_mechanism.json`): the formula  $|V_{us}| \approx \varepsilon_1^{\alpha_d}$  gives a 1.9% match while applying  $\alpha_d$  to zero-charge elements degrades the prediction, confirming the mechanism is  $\Delta a$ -specific.

The algebraic structure of Eqs. (6)–(5) is now fully Lean-certified in the module `UgpLean.MassRelations.CKMMixing` (11 theorems, zero sorry): `cabibbo_effective_charge` proves  $\Delta a_{\text{eff}} = \alpha_d$  from the GUT coefficient; `cabibbo_charge_from_GUT` establishes the GUT group-theory origin  $\alpha_d = 1 + \text{rank}(\text{SU}(5))/N_c^2$ ; and `cabibbo_vev_formula` certifies  $|V_{us}|_{\text{CDM}} = \varepsilon_1^{\alpha_d}$ . The algebraic FN diagonalization bridge is also certified: `fn_vv_correction_additive` proves that  $(\alpha_d - 1)$  shifts the bare FN charge *additively*, and `fn_diagonalization_vv_bridge` certifies the log formula  $\Delta a_{\text{eff}} \cdot \log \varepsilon_1 = -13\pi/27$  (both zero sorry). The remaining open physical step is the full  $2 \times 2$  SVD identification,  $|(U_{uL}^\dagger U_{dL})_{12}| = \varepsilon_1^{\alpha_d} \cdot (1 + \mathcal{O}(\varepsilon_1^2))$ , which requires Weyl–Courant perturbation theory on  $Y_d Y_d^\dagger$ ; once proved, this would elevate Eq. (5) from a [C] structural hypothesis to a [T] Lean-certified result.

The companion paper [11] takes a different structural route to the Wolfenstein parameters, deriving  $\lambda$  directly from the GTE generation and family counts:  $\lambda = N_{\text{gen}}^2 / (2^{N_{\text{gen}}} N_{\text{fam}}) = 9/40 = 0.22500$ , achieving exact agreement with the PDG value ( $0.0\sigma$ ) without the VV correction mechanism used here. The two routes are complementary: the approach in this paper provides the cyclotomic-geometric mechanism (VV GUT coefficient) underlying the 1.9% estimate  $\varepsilon_1^{\alpha_d} \approx 0.2203$ , while [11] identifies the combinatorial formula that closes the remaining gap.

#### Open Problem 4 — PMNS + $\delta_{\text{CP}}$

Construct the PMNS matrix from the FN-doubled charged-lepton sector and anti-doubled  $\nu_R$  sector; predict  $\delta_{\text{CP}}$  for DUNE Run-1.



## 7 Conclusions

We have presented two structural mass relations reducing the nine charged-fermion masses to two scalar inputs, and identified their cyclotomic-12 geometric origin:

1. The TT coefficient  $\alpha = \pi/6$  is the  $A_2$  Weyl-chamber half-opening angle, Lean-proved (`angle_alpha1_omega1_eq_pi_div_six`). The  $2^9$  structure is Lean-proved via a binary cascade (`cascadeState_eq_TT`). The full TT formula is structurally realised by a two-flavon FN UV completion with VEVs at Lean-proved potential minima (`fn_vevs_are_potential_minima`).
2. The VV formula is a genuine structural regularity (null density  $\leq 10^{-5}$ ). Its coefficient values  $(13/9, -7/6, -5/14)$  are structurally derived from  $SU(5)/SO(10)$  GUT group theory:  $\alpha_d = 1 + \text{rank}(SU(5))/N_c^2$ ,  $\beta_d = -(1 + Y_{QL})$ ,  $\gamma_d = -\dim(45_{SU(5)})/\dim(126_{SO(10)})$ . Machine-checked: `VV_from_GUT_group_theory` (zero sorry). The classification is [E/C].
3. Combined with the Koide closed form [10], all nine charged-fermion masses are determined from two inputs, end-to-end machine-checked in `UgpLean.MassRelations.PhysicalMasses`.
4. The cyclotomic-12 atom family  $\{\pi/6, \pi/8, \cos(\pi/12), \sqrt{3}\}$  unifies the TT and Koide structures in a single Lie-geometric framework.

The Lean 4 formalisation (11 modules, zero sorry, standard Mathlib axioms) demonstrates end-to-end machine-auditability of the prediction chain and sets a template for future fermion-mass structural identifications.

## Code Availability

All computational artifacts are in the companion repository [13] under `papers/01_SM/canonical_run/`. Key standalone scripts for this paper: `comp_p01_TT_up_lepton_cyclotomic_identity.py` (TT identity verification and null-density test), `comp_p01_VV_down_linked_to_up_lepton.py` (VV identity verification), `comp_p01_XX_gut_structure_search.py` (structural search over GUT representation-theoretic atoms), `comp_p01_EBF_14_vv_rg_flow.py` and `comp_p01_EBF_15_vv_gj_su5_full.py` (one-loop and full GJ/SU(5) RGE tests), `comp_p01_EBF_16_vv_gut_group_theory.py` (exact Weyl-formula verification for  $A_4 = SU(5)$  and  $D_5 = SO(10)$ ). Each script is independently reproducible with Python 3.9+ and numpy (plus sympy for the group-theory scripts). Lean formalisation: `ugp-lean` [12] (`lake build` to verify, Lean 4.29.0-rc6 / Mathlib 4.29.0-rc6). See `PROVENANCE.md` and `REPRODUCE.md` in the companion repository for SHA-256 hashes and step-by-step instructions.

## Data Availability

All data used are PDG 2022 central values, publicly available at <https://pdg.lbl.gov/>.

## A Complete Lean Module Listing

The following modules in `UgpLean.MassRelations` are relevant to this paper. All build clean under `lake build`; all zero sorry.

```

UgpLean/MassRelations/
  SU3FlavorCartan.lean      -- A2 Weyl-chamber angle pi/6
  BinaryCascade.lean        -- Binary cascade -> TT closed form
  FroggattNielsen.lean      -- FN UV completion; beta = pi/8
  CartanFlavonPotential.lean -- Z6 x Z16 potential global minima
  Z2OrbifoldDepth.lean      -- Doubled FN charges from Z2-tree depth
  HeavyFermionTower.lean     -- EFT duality: FN = heavy-fermion tower
  UpLeptonCyclotomic.lean    -- TT scaffolding; beta = alpha/C2(SU3)
  DownRational.lean          -- VV rational form; N_c formulas + GUT group theory
  ClebschGordan.lean         -- CG coefficient structural anchors
  CKMMixing.lean             -- CDM: cabibbo_effective_charge, cabibbo_charge_from_GUT,
                                -- cabibbo_vev_formula; zero sorry
  KoideClosedForm.lean       -- Koide m_tau prediction (see companion paper)
  KoideAngle.lean            -- theta = (N_c^2 - 1)/(4 N_c^2) = 2/9 from N_c = 3;
                                -- N_c structural chain (delta, b1, a_top, strand);
                                -- nu seesaw exponent 29/9 (three decompositions);
                                -- dim(126_S010) = 2 N_c^2 delta cross-identity
  KoideNewtonFlow.lean       -- S3-equivariant Newton flow
  LeptonMassPrediction.lean   -- Full lepton mass pipeline from (m_e, theta)
  ClaimCBridge.lean          -- Formal TT cascade: (pi/6) * 2^g + pi/8
  PhysicalMasses.lean         -- End-to-end capstone (TT + VV + Koide)

UgpLean/ElegantKernel/Unconditional/
  KGenFullClosure.lean       -- k_gen = phi * cos(pi/10); pentagon-hexagon bridge

```

Repository: <https://github.com/novaspiack/ugp-lean> (build: lake build). Zenodo DOI: [12].

## B Derivation of the $A_2$ Weyl-Chamber Angle

We give an explicit two-dimensional construction of eq. (4).

The  $A_2$  root system in the Cartan plane  $\mathbb{R}^2$  has simple roots  $\alpha_1 = (1, 0)$  and  $\alpha_2 = (-1/2, \sqrt{3}/2)$  (at  $120^\circ$ ). The fundamental weight  $\omega_1$  dual to  $\alpha_1$  satisfies  $\langle \omega_1, \alpha_1^\vee \rangle = 1$  and  $\langle \omega_1, \alpha_2^\vee \rangle = 0$ , where  $\alpha^\vee = 2\alpha/\langle \alpha, \alpha \rangle$ . In Cartesian coordinates:  $\omega_1 = (1/2, \sqrt{3}/6)$  (one can verify  $\langle \omega_1, \alpha_1 \rangle / |\alpha_1|^2 = 1/2$  and  $\langle \omega_1, \alpha_2 \rangle / |\alpha_2|^2 = 0$ ).

The angle between  $\alpha_1 = (1, 0)$  and  $\omega_1 = (1/2, \sqrt{3}/6)$  is

$$\theta = \arctan\left(\frac{\sqrt{3}/6}{1/2}\right) = \arctan\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}.$$

The fundamental Weyl chamber is bounded by the half-planes  $\langle v, \alpha_i \rangle \geq 0$  for  $i = 1, 2$ , which in polar coordinates corresponds to angles  $\theta \in [0, \pi/3]$ . The interior bisector is at  $\theta = \pi/6$ , i.e.,  $\omega_1$  lies on the chamber bisector.

This is the content of `angle_alpha1_omega1_eq_pi_div_six` and `omega1_in_weyl_chamber_interior` (both in `UgpLean.MassRelations.SU3FlavorCartan`).

## C Pre-Registered Basis Definitions for SC-JJJ, SC-KKK, SC-LLL

### SC-JJJ: GUT-representation basis (96 atoms)

Basis SHA-256 (committed before scan):

644a7f90bc9682db672a8d8920df80a2c481e232c10f44c316a1321b626e8899.

Atom classes: classical Lie-group ranks  $\{\text{rank}(\text{SU}(n))\}_{n=2}^8$ ,  $\{\text{rank}(\text{SO}(n))\}_{n=4}^{10}$ ,  $\{\text{rank}(E_n)\}_{n=6}^8$ ,  $\text{rank}(F_4)$ ,  $\text{rank}(G_2)$ ; representation dimensions of  $\text{SU}(3)$ ,  $\text{SU}(5)$ ,  $\text{SO}(10)$  including  $\dim(45_{\text{SU}(5)}) = 45$ ,  $\dim(126_{\text{SO}(10)}) = 126$ ; SM hypercharges  $Y_Q = 1/6$ ,  $Y_u = 2/3$ ,  $Y_d = -1/3$ ,  $Y_L = -1/2$ ,  $Y_e = -1$ ,  $Y_H = 1/2$ ; gauge-boson counts  $\{1, 3, 8, 4, 12\}$ ; small rationals  $\{1/n\}_{n=1}^{14}$ . DL  $\leq 3$ : 603,840 expressions.

### SC-KKK: FN integer-charge derivation (analytical)

Pre-commit SHA-256 (artifact `comp_p01_KKK_vv_rg_derivation.json`):

1c4bb21b387519209b912d5bf0576a54b1e469c6ec25ce2eecfd8975be622698.

Derived analytically: the FN-doubled framework predicts down-sector charges  $q_{d,g}^{(1)} = -(7/6) \cdot 2^{g-1}$  and  $q_{d,g}^{(2)} = -13/9$ , both non-integer. Seven integer-charge candidates tested computationally; all fail.

### SC-LLL: Discrete-flavor basis (97 atoms)

Basis SHA-256 (committed before scan):

dc210d1240f9d8f8d3438fdf0e2443016f2e698f3d4d857a516360fd9861f180.

Discrete-symmetry groups from [6]:  $A_4$ ,  $S_4$ ,  $T'$ ,  $A_5$ ,  $\Delta(27)$ ,  $\Delta(54)$ ,  $\Sigma(168)$ ,  $\Sigma(216)$ ,  $\Sigma(72)$ ,  $\Sigma(36)$ ,  $D_3$ – $D_8$ ; subgroup indices and representation dimensions; cyclotomic cosines  $\cos(2\pi k/n)$  for  $n \in \{3, 4, 5, 6, 7, 8\}$ . DL  $\leq 3$ : 610,421 expressions.

## D SHA-256 Artifact Table

Table 5: SHA-256 hashes of all computational artifacts used in this paper. Artifacts are in `papers/01_SM/canonical_run/`.

| Artifact | File                               | SHA-256 (first 16 hex chars) |
|----------|------------------------------------|------------------------------|
| SC-TT    | <code>comp_p01_TT_....json</code>  | d8227f9d651b3d95             |
| SC-VV    | <code>comp_p01_VV_....json</code>  | 10f1bb1e099d444d             |
| SC-EEE   | <code>comp_p01_EEE_....json</code> | 120fd2aa2d2270f1             |
| SC-JJJ   | <code>comp_p01_JJJ_....json</code> | f82288b4e6816cfe             |
| SC-KKK   | <code>comp_p01_KKK_....json</code> | e9956111050db7b6             |
| SC-LLL   | <code>comp_p01_LLL_....json</code> | db997f06ff06160f             |
| SC-BBB   | <code>comp_p01_BBB_....json</code> | ef33c48fa938e3fa             |

## E Six $\beta$ -Free Inter-Generational Identities

Table 6: The six  $\beta$ -free inter-generational identities derived from the TT formula by taking differences  $R_g - R_{g'}$ . No free parameters.

| Identity                                                  | Predicted | PDG    | Residual (%) | Note         |
|-----------------------------------------------------------|-----------|--------|--------------|--------------|
| $\log(m_c/m_e) - \log(m_u/m_e) = (\pi/6)(2^2 - 2^1)$      | 5.890     | 5.868  | +0.37        |              |
| $\log(m_t/m_\tau) - \log(m_c/m_\mu) = (\pi/6)(2^3 - 2^2)$ | 5.243     | 5.250  | −0.13        |              |
| $\log(m_t/m_\tau) - \log(m_u/m_e) = (\pi/6)(2^3 - 2^1)$   | 11.133    | 11.118 | +0.14        |              |
| $\log(m_c m_e / m_u m_\mu)$                               | $\pi/3$   | 1.044  | −0.03        |              |
| $\log(m_t m_\mu / m_c m_\tau)$                            | $\pi/3$   | 1.046  | −0.16        |              |
| $\log(m_t m_e / m_u m_\tau)$                              | $\pi$     | 3.134  | −0.23        | Gelfond-type |

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